Exam Date & Time: 09-Dec-2023 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

III SEMESTER B.TECH END SEMESTER EXAMINATIONS, DECEMBER 2023

DATA ANALYTICS [DSE 2121]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Use the transactional database given in Table 1. With 33.34% support threshold build a frequent pattern tree (FP-Tree). Show for each transaction how the tree evolves. Build Fp-Growth to discover the frequent itemsets from this FP-tree.

A)

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

(5)

Table 1. Transaction itemset

B) Consider the following set of exam scores for a group of students: 85, 92, 78, 89, 95, 87, 78, 90, 88.

I. Calculate the mean, median, and mode of the exam scores.

II. Interpret the results in the context of student performance.

III. Analyse a real-world scenario where central tendency is crucial.

(3)

C) You are tasked with creating a visual representation of quarterly sales data for a company. Apply Edward Tufte's principles of data visualization to design a clear and effective chart.

I. Identify an appropriate type of chart to represent quarterly sales data.

II. Apply at least two of Tufte's principles to enhance the clarity and effectiveness of your chart.

(2)

2) Build three clusters using the below eight points:

(5)

A) A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)

Consider the Initial cluster centres as: A1(2, 10), A4(5, 8) and A7(1, 2).

B) A survey was conducted in the city. Table 2 provides the sample data containing a person's age and their corresponding income.

Age	25	30	36	43
Income	30000	44000	52000	7000

(3)

Table 2. person's age and income.

Find out whether the increase in age has an effect on income using the correlation coefficient.

C) Assume that you have collected data on the hours students spend studying per week and their corresponding grades in a particular course. Based on the dataset, Analyse involved bivariate analysis and Justify it's significance in the context of studying the relationship between study hours and grades. (2)

3) Manufacturers of golf balls always seem to be claiming that their ball goes the farthest. A writer for a sports magazine decided to conduct an impartial test. She randomly selected 20 golf professionals and then randomly assigned four golfers to each of five

A) brands. Each golfer drove the assigned brand of ball. The driving distances, in yards, are displayed in Table 3.

Brand 1	Brand 2	Brand 3	Brand 4	Brand 5
286	279	270	284	281
276	277	262	271	293
281	284	277	269	276
274	288	280	275	292

(5)

Table 3. Driving distance

Preliminary data analyses indicate that the independent samples come from normal populations with equal standard deviations. Do the data provide sufficient evidence to conclude that a difference exists in mean weekly earnings among nonsupervisory workers in the five industries? Perform the required hypothesis test using $\alpha = 0.05$.

B) A poker-dealing machine is supposed to deal cards at random, as if from an infinite deck. In a test, you counted 1600 cards, and observed the following:

Spades	404

(3)

Hearts	420
Diamonds	400
Clubs	376

Does the machine have an equal probability of dealing a card from any of the four suits, or do the observed discrepancies suggest that the outcomes having random chance?

C) *"Assume that you are a practicing data analyst leading a team through the initial phases of a data analytics project for a retail company. The company wants to optimize its inventory management system by leveraging data analytics to predict demand patterns. Your team is tasked with analysing the problem definition stages to ensure a clear understanding of the project goals."* (2)

As the lead data analyst, analyse the problem definition stages in the context of data analytics.

4) Table 4 provides the database with 5 transactions and a minimum support threshold of 60% and a minimum confidence threshold of 80%,

A) I. Find all frequent itemsets using Apriori algorithm.

II. List all strong association rules that contain "A" in the antecedent (Constraint). (5)

TID	Transaction
T1	{A, B, C, D, E, F}
T2	{B, C, D, E, F, G}
T3	{A, D, E, H}
T4	{A, D, F, I, J}
T5	{B, D, E, K}

Table 4. Transaction List

B) Suppose you start up a company that has developed a drug that is supposed to increase IQ. You know that the standard deviation of IQ in the general population is 15. You test your drug on 36 patients and obtain a mean IQ of 97.65. Using an alpha value of 0.05, is this IQ significantly different than the population mean of 100? (3)

C) Identify, how user-based collaborative filtering and item-based collaborative filtering differ in terms of their approach and addressing the cold start problem? (2)

5) Apply single-link agglomerative clustering to form clusters for the provided samples:

A1=(2,10), A2=(2,5), A3=(8,4), A4=(5,8), A5=(7,5), A6=(6,4), A7=(1,2), A8=(4,9). (5)

A)

B) Given below is a set table that contains some items and the user who have rated those items. The rating is explicit and is on a scale of 1 to 5. Each entry in the table denotes the rating given by a ith user to a jth Item. Find the missing ratings for the respective user.

User/Item	Item_1	Item_2	Item_3
User_1	2	-	3
User_2	5	2	-
User_3	3	3	1
User_4	-	2	2

(3)

C) *"A marketing analytics firm preparing a comprehensive report for a client that involves presenting complex data insights visually and informatively. Your task is to analyse and explain the critical role of graphical properties in conveying information effectively to the client."*

(2)

As a specialist, analyse the significance of graphical properties in information visualization.

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